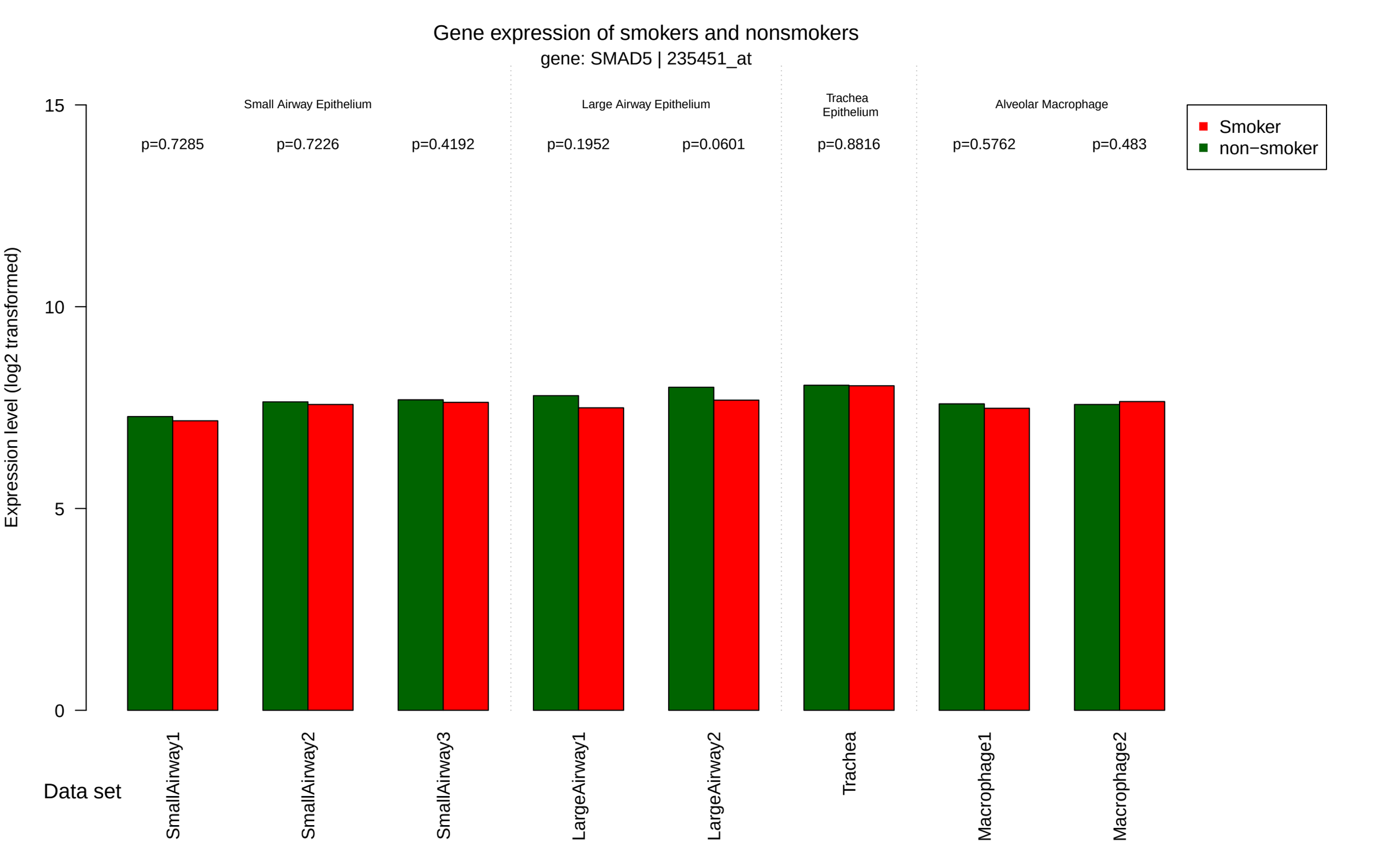
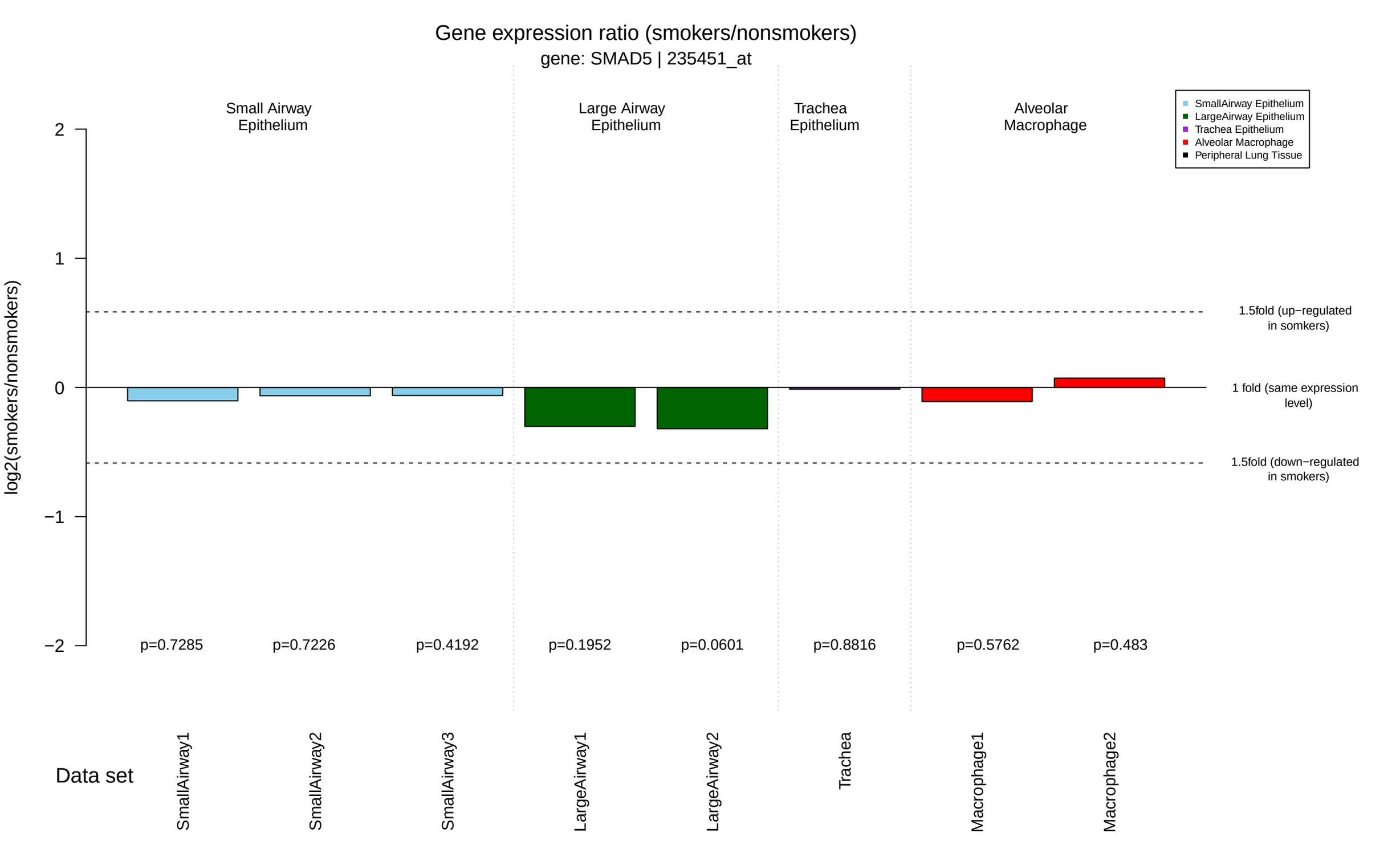
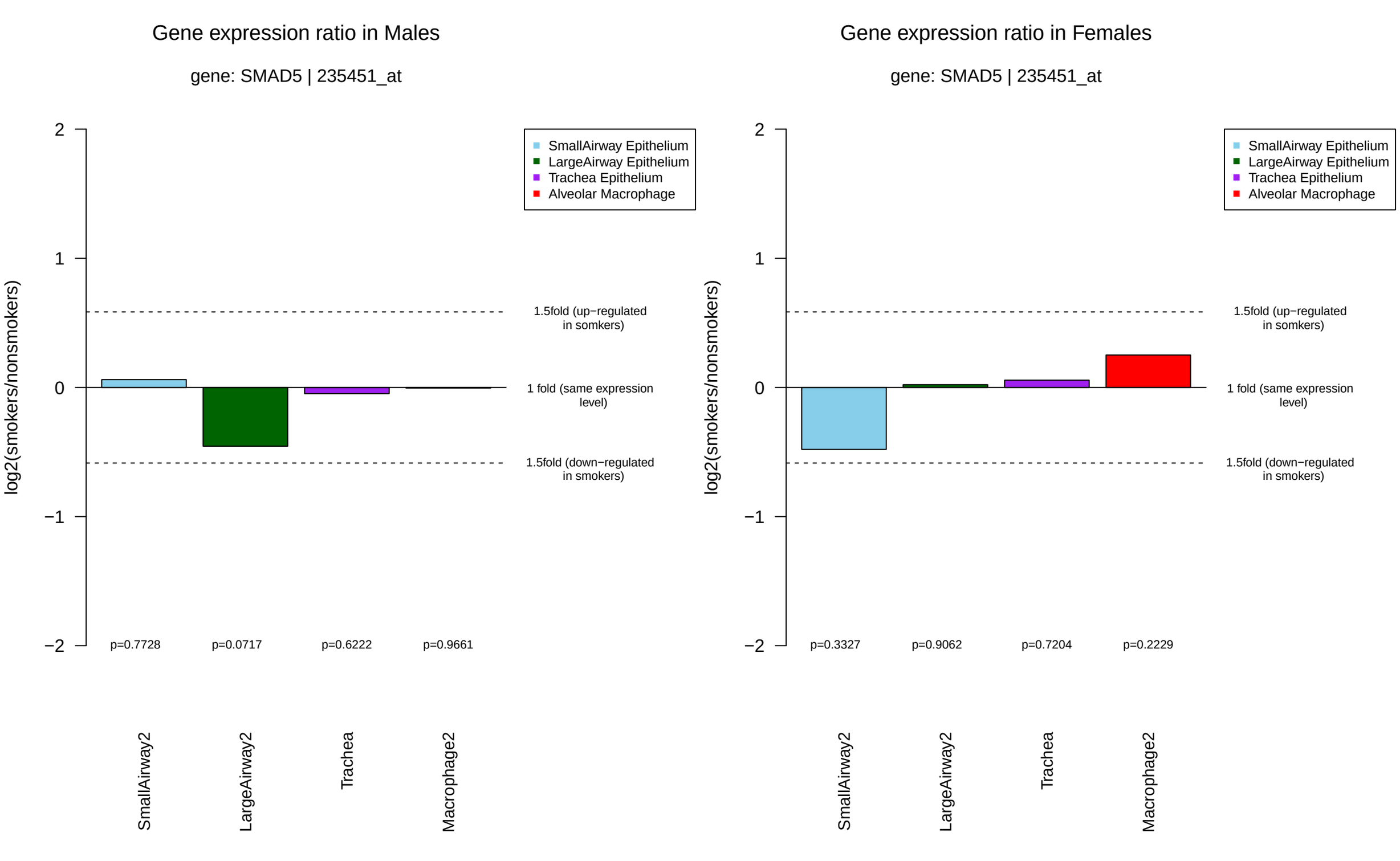
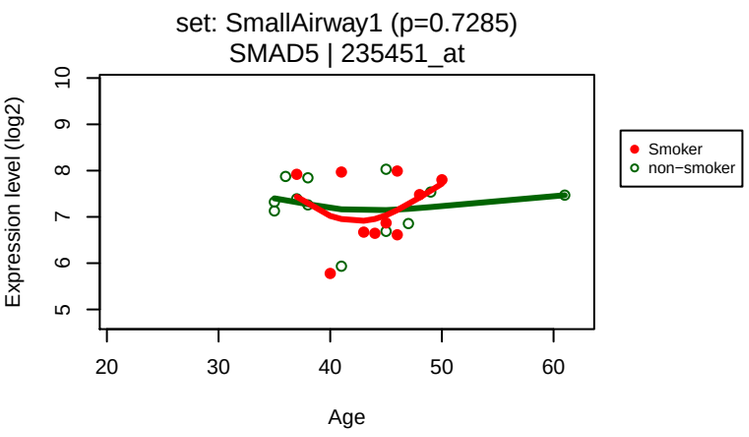
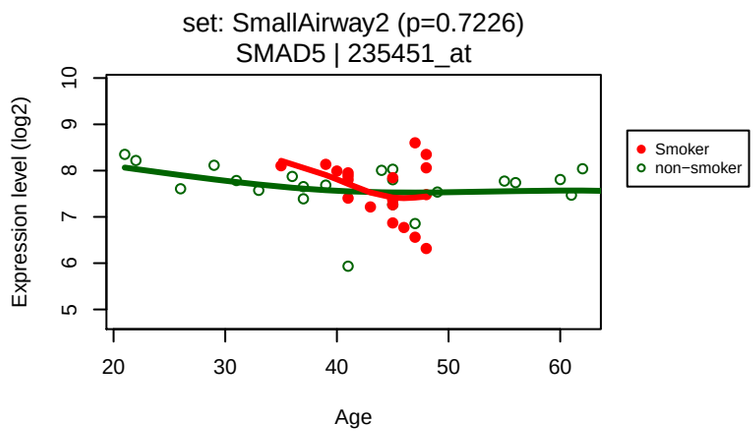
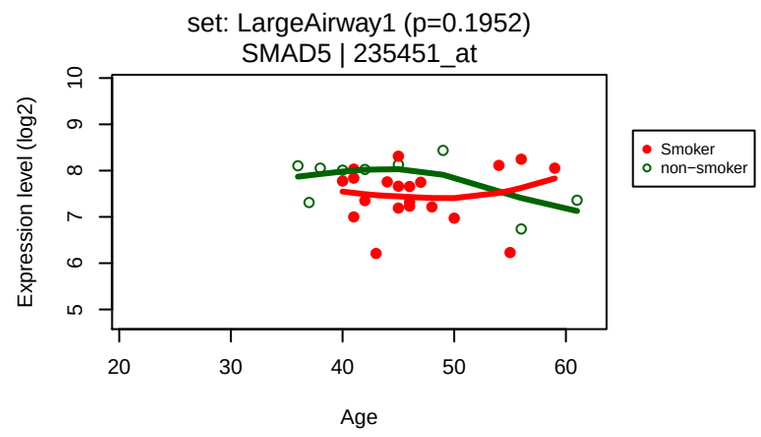

Gene expression of smokers and nonsmokers
gene: SMAD5 | 235451_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7285
p=0.7226
p=0.4192
p=0.1952
p=0.0601
p=0.8816
p=0.5762
p=0.483
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

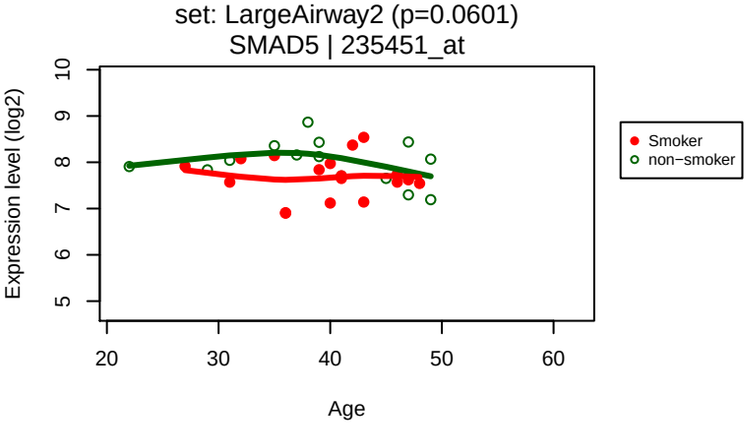

Gene expression ratio (smokers/nonsmokers)
gene: SMAD5 | 235451_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7285
p=0.7226
p=0.4192
p=0.1952
p=0.0601
p=0.8816
p=0.5762
p=0.483
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

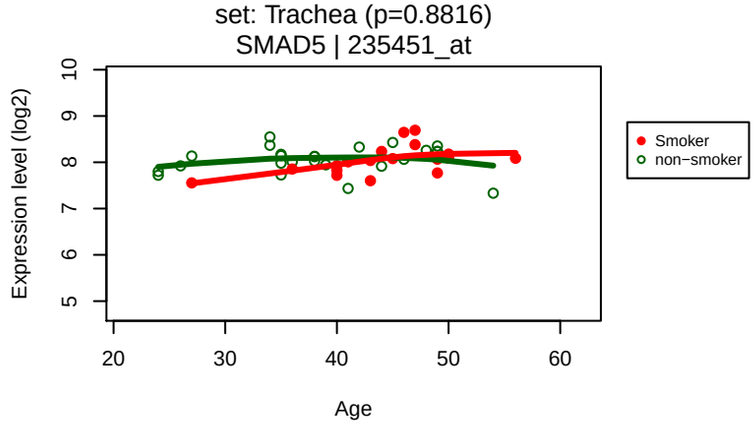

Gene expression ratio in Males
gene: SMAD5 | 235451_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7728
p=0.0717
p=0.6222
p=0.9661
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD5 | 235451_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3327
p=0.9062
p=0.7204
p=0.2229
SmallAirway2
LargeAirway2
Trachea
Macrophage2

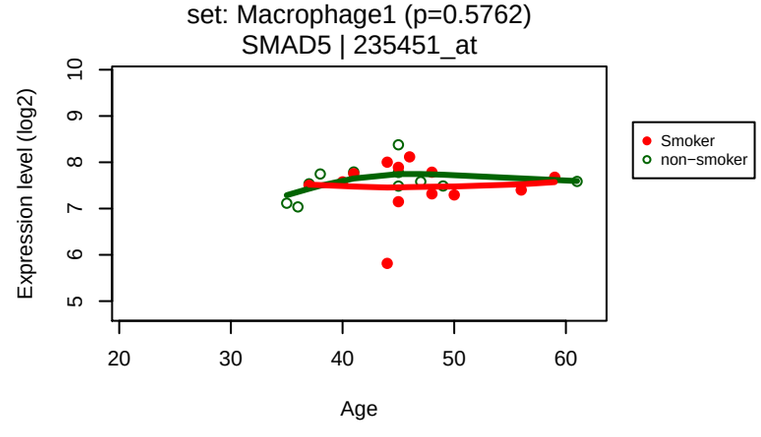

set: SmallAirway1 (p=0.7285)
SMAD5 | 235451_at
Expression level (log2)
Age
Smoker
non−smoker

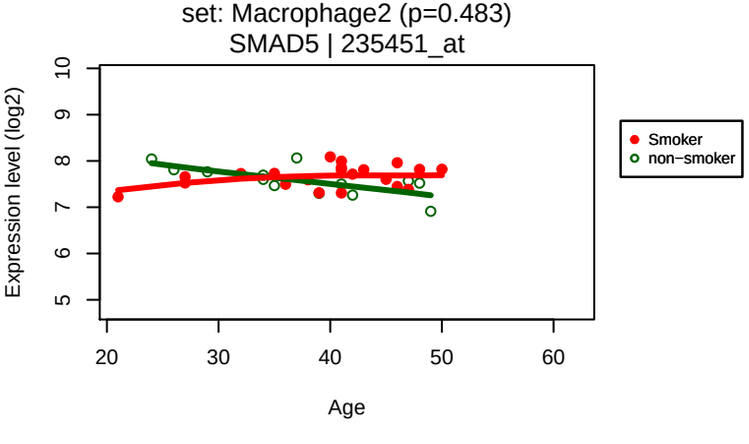

set: SmallAirway2 (p=0.7226)
SMAD5 | 235451_at
Expression level (log2)
Age
Smoker
non−smoker

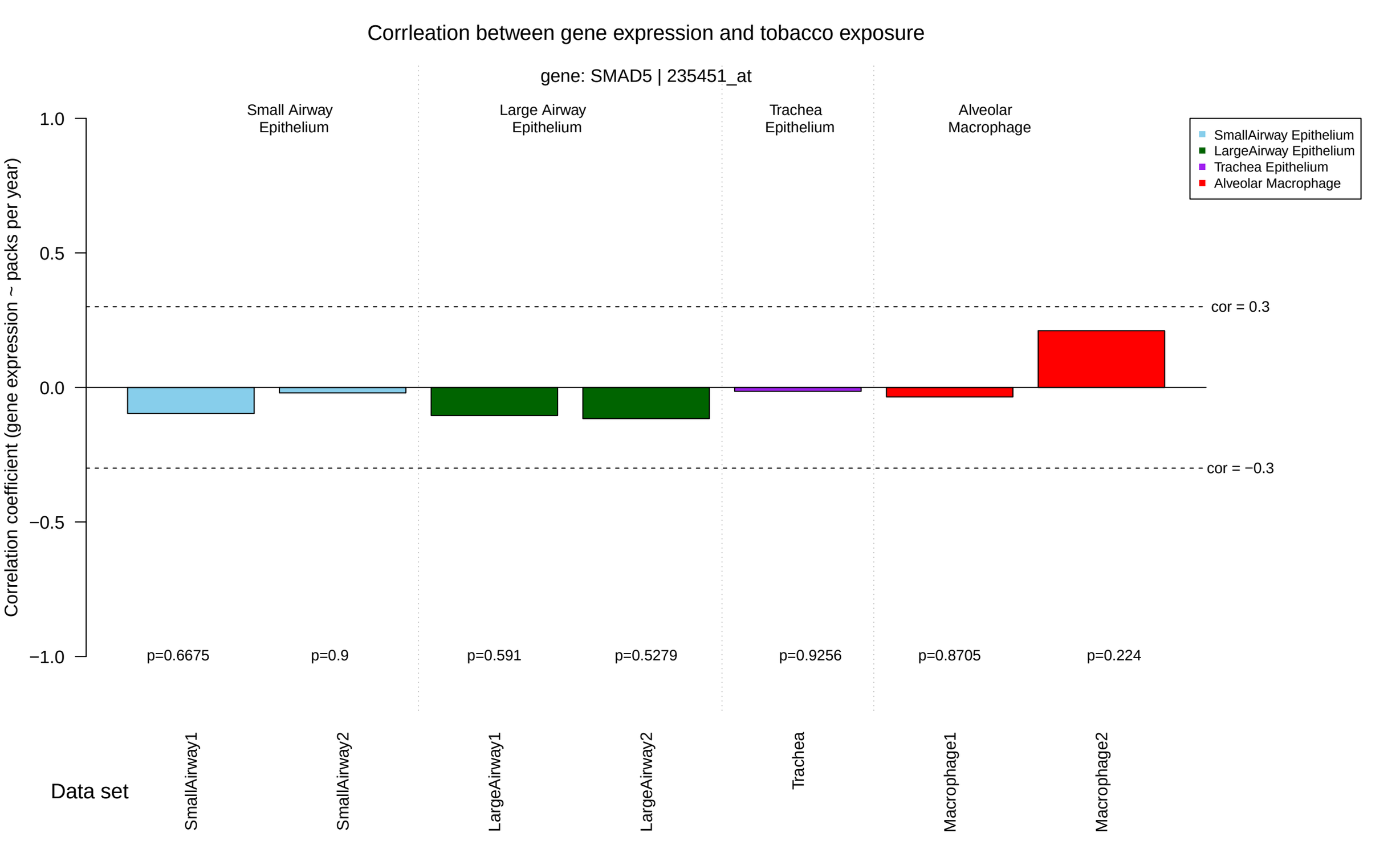

set: LargeAirway1 (p=0.1952)
SMAD5 | 235451_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0601)
SMAD5 | 235451_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8816)
SMAD5 | 235451_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5762)
SMAD5 | 235451_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.483)
SMAD5 | 235451_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD5 | 235451_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6675
p=0.9
p=0.591
p=0.5279
p=0.9256
p=0.8705
p=0.224
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2